

Critical Study Summary: Kampmann et al. (2023) Maternal RSVpreF Vaccine Trial

Study Design and Population

Phase 3, randomized, double-blind, placebo-controlled maternal vaccination trial conducted across 18 countries. Pregnant women at 24–36 weeks' gestation were randomized 1:1 to receive RSVpreF vaccine or placebo. Infant efficacy analyses included over 7,100 infants with follow-up from birth.

Follow-up and Time Scale

Infants were followed from birth with active surveillance for RSV-associated outcomes. Efficacy was reported at multiple postnatal time points (90, 120, 150, and 180 days), allowing assessment of temporal changes in protection.

Primary Outcomes and Effect Measures

Primary endpoints were medically attended severe RSV-associated lower respiratory tract infection (LRTI) and medically attended RSV-LRTI of any severity. Vaccine efficacy was estimated using relative risks within prespecified time windows.

Key Findings Relevant to Waning

Vaccine efficacy against severe RSV-LRTI declined from approximately 82% at 90 days to about 69% at 180 days after birth, indicating measurable waning over the first six months of life.

Graphical Data for Reconstruction

Cumulative incidence curves stratified by treatment group are presented with numbers at risk and event times through 180 days, providing suitable inputs for individual patient data reconstruction.

Strengths for Waning Analysis

Large multinational randomized design, prespecified time windows, laboratory-confirmed outcomes, and clear censoring rules support secondary time-to-event modeling.

Limitations for Waning Analysis

Primary reporting used risk ratios rather than hazard models; interim analysis and limited long-term follow-up may affect estimation of later waning dynamics.